Q P Code:-25900507

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMAT201 - BUILDING BLOCKS FOR HIGHER MATHEMATICS

(2024 ADMISSION ONWARDS)

Scheme For Valuation

Duration: 2 Hours Maximum Marks: 70

Part A

(Short Answer Questions)
Answer any 5 questions
Each question carries 2 marks

Question 1 What is meant by a direct proof of a conditional statement $p \to q$? **[K] / [C01]**

- Correct Definition ... 2 Marks

Question 2 Let $\mathcal{L}$ denote the set of all lines in a plane. For any two lines l_1 and l_2, we write if l_1 is perpendicular to l_2. Check whether "perpendicular to" is a relation on $\mathcal{L}$.
[An] / [C02]

- Yes, because set of all ordered pairs of perpendicular lines is a subset of $\mathcal{L} \times \mathcal{L}$ ... 2 Marks

Question 3 Consider the poset $(\mathbf{Z} \times \mathbf{Z}, \preceq)$ where $\preceq$ is the lexicographic ordering constructed from the usual $\preceq$ relation on $\mathbf{Z}$. Determine whether

(a) $(3, 5) \prec (4, 8)$

(b) $(3, 8) \prec (4, 5)$

(c) $(4, 9) \prec (4, 11)$

(d) $(5, 3) \prec (4, 4)$

[U] / [C03]

- $(a_1, a_2) \preceq (b_1, b_2)$ either if $a_1 < b_1$ or if both $a_1 = b_1$ and $a_2 < b_2$
- $(3, 5) \prec (4, 8)$ is True, since $3 < 4$ ... 1 Mark

- $(3, 8) \prec (4, 5)$ is True, since $3 < 4$ ... 1 Mark
- $(4, 9) \prec (4, 11)$ is True, since $4 = 4$ and $9 < 11$ ... 1 Mark
- $(5, 3) \prec (4, 4)$ is False, since $5 \not< 4$ ... 1 Mark

Total Marks should not Exceed 2

Question 4 Construct a partition on the set $\{a, b, c, e, f, b, i\}$. **[A] / [C04]**

- For any correct partition on the set $\{a, b, c, e, f, b, i\}$ ... 2 Marks
  Here the element b is repeated, give full marks to any correct partition with or without the repeated element

Question 5 Find the unit tangent vector of the plane curve $\mathbf{r}(t) = t^2\mathbf{i} + e^t\mathbf{j}$. **[A] / [C04]**

- $\overline{v}(t) = 2t\,\mathbf{i} + e^t\,\mathbf{j}$ ... 1 Mark
- $|\overline{v}(t)| = \sqrt{(2t)^2 + (e^t)^2}$ ... 1 Mark
- $\mathbf{T}(t) = \dfrac{\overline{v}(t)}{|\overline{v}(t)|} = \dfrac{2t\,\mathbf{i} + e^t\,\mathbf{j}}{\sqrt{4t^2 + e^{2t}}}$. ... 1 Mark

Total Marks should not Exceed 2

Question 6 Find principal unit normal $\mathbf{N}$ for the space curve whose unit tangent vector $\mathbf{T}$ is given by $\mathbf{T} = \left(\dfrac{1}{\sqrt{2}}\sin 4t\right)\mathbf{i} + \left(-\dfrac{1}{\sqrt{2}}\cos 4t\right)\mathbf{j} + \dfrac{1}{\sqrt{2}}\mathbf{k}$. **[A] / [C04]**

- $\dfrac{d\mathbf{T}}{dt} = 2\sqrt{2}\big(\cos(4t)\,\mathbf{i} + \sin(4t)\,\mathbf{j}\big)$ ... 1 Mark
- $\left|\dfrac{d\mathbf{T}}{dt}\right| = 2\sqrt{2}$ ... 1 Mark
- $\mathbf{N}(t) = \dfrac{\dfrac{d\mathbf{T}}{dt}}{\left|\dfrac{d\mathbf{T}}{dt}\right|} = \cos(4t)\,\mathbf{i} + \sin(4t)\,\mathbf{j}$. ... 1 Mark

Total Marks should not Exceed 2

Question 7 Test whether the differential form $(x + y)\,dx + y\,dy + (y + z)\,dz$ is exact. **[K] / [CO5]**

- Here $M = x + y, N = y, P = y + z$ ... 1 Mark
- $\dfrac{\partial P}{\partial y} = 1 \quad \dfrac{\partial N}{\partial z} = 0$, so $\dfrac{\partial P}{\partial y} \neq \dfrac{\partial N}{\partial z}$
  So it is not exact ... 1 Mark

OR

- $\frac{\partial N}{\partial x} = 0 \quad \frac{\partial M}{\partial y} = 1$, so $\frac{\partial N}{\partial x} \neq \frac{\partial M}{\partial y}$
  So it is not exact ... 1 Mark

Question 8 Find the curl of $\mathbf{F}(x, y, z) = xz\,\hat{i} + xy\,\hat{j} + 3xz\,\hat{k}$. **[A] / [CO5]**

- Curl $= \nabla \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & xy & 3xz \end{vmatrix}$ ... 1 Mark
- Calculations ... 1 Mark
- Final answer **Curl**$=0\,\hat{i} + (x - 3z)\,\hat{j} + y\,\hat{k}$. ... 1 Mark

Total Marks should not Exceed 2

Part B
(Short Essay Questions)
Answer any 5 questions
Each question carries 6 marks

Question 9 Prove that if x is irrational, then $\frac{1}{x}$ is irrational. **[A] / [CO1]**

- For taking $\frac{1}{x}$ as rational ... 3 Marks
- For proving x is rational ... 3 Marks

Question 10 (a) What do you mean by "circular reasoning"? **[A] / [CO1]**

- Correct Definition ... 2 Marks

(b) Is the following argument correct?

It supposedly shows that n is an even integer whenever n^2 is an even integer.
Suppose that n^2 is even.
Then $n^2 = 2k$ for some integer k.
Let $n = 2l$ for some integer l.
This shows that n is even.

- This argument is incorrect. The statement "let n = 2l for some integer l" occurs in the proof. No argument has been given to show that n can be written as 2l for some integer l. This is circular reasoning because this statement is equivalent to the statement being proved, namely, "n is even."
  ... 4 Marks

Question 11 Determine whether the poset $(\mathcal{P}(S), \subseteq)$ is a lattice for any set S. **[U] / [CO3]**

- Define the conditions to prove $(\mathcal{P}(S), \subseteq)$ is a lattice ... 1 Mark

- Arguments to show $A \cap B$ is a lower bound of A and B, for any $A, B \in \mathcal{P}(S)$ ... 3 Marks
- Arguments to show $A \cup B$ is an upper bound of A and B, for any $A, B \in \mathcal{P}(S)$ ... 3 Marks
- Conclusion ... 1 Mark

Total Marks should not Exceed 6

Question 12 Draw the Hasse diagram representing the partial ordering $\{(a, b) \mid a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$. **[U] / [CO3]**

- Correct Diagram ... 6 Marks
- Diagram with minor mistakes ... 3 Marks

Question 13 Evaluate the speed and acceleration of the position vector $\mathbf{r}(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j} + \sin(2t)\,\mathbf{k}$ at $t = \dfrac{\pi}{2}$. **[A] / [CO4]**

- Velocity: $\bar{\mathbf{v}}(t) = \dfrac{d\mathbf{r}}{dt} = -\sin t\,\mathbf{i} + \cos t\,\mathbf{j} + 2\cos(2t)\,\mathbf{k}$. ... 2 Marks
- Acceleration: $\mathbf{a}(t) = \dfrac{d\mathbf{v}}{dt} = -\cos t\,\mathbf{i} - \sin t\,\mathbf{j} - 4\sin(2t)\,\mathbf{k}$. ... 2 Marks
- $\mathbf{a}\left(\frac{\pi}{2}\right) = -\mathbf{j}$. ... 1 Mark
- Speed: $|\bar{\mathbf{v}}(t)| = \sqrt{(-\sin t)^2 + (\cos t)^2 + (2\cos(2t))^2}$, ... 2 Marks
- $|\mathbf{v}\left(\frac{\pi}{2}\right)| = \sqrt{1^2 + 0^2 + (-2)^2} = \sqrt{5}$. ... 2 Marks

Total Marks should not Exceed 6

Question 14 Find ∇f if $f(x, y, z) = 2z^3 - 3(x^2 + y^2)z + \tan^{-1}(xz)$, at $(1, 1, 1)$. **[U] / [CO4]**

- For finding $\dfrac{\partial f}{\partial x}$, $\dfrac{\partial f}{\partial y}$, $\dfrac{\partial f}{\partial z}$ ... 3Marks
- Evaluate partial derivatives at $(x, y, z) = (1, 1, 1)$ ... 3 Marks
- Final Answer $\nabla f(1, 1, 1) = -\dfrac{11}{2}i - 6j + \dfrac{1}{2}k$ ... 1 Mark

Total Marks should not Exceed 6

Question 15 A fluid's velocity field is $\vec{F} = -4xy\,\mathbf{i} + 8y\,\mathbf{j} + 2\,\mathbf{k}$. Find the flow along the curve $\vec{\mathbf{r}}(t) = t\,\mathbf{i} + t^2\,\mathbf{j} + \mathbf{k},\ 0 \le t \le 2$. **[An] / [CO5]**

- Compute the derivative of the position vector $\vec{r}\,'(t) = \dfrac{d\vec{r}}{dt} = \mathbf{i} + 2t\,\mathbf{j} + 0\,\mathbf{k}$ ... 2 Marks
- Evaluate $\vec{F}$ along the curve $\vec{F}(\vec{r}(t)) = -4t^3 i + 8t^2 j + 2k$ ... 1 Mark
- Compute the dot product $\vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t) = 12t^3$. ... 2 Marks
- Evaluate the line integral $\displaystyle\int_C \vec{F} \cdot d\vec{r} = \int_0^2 12t^3\,dt = 48$ ... 2 Marks

Total Marks should not Exceed 6

Question 16 Find a potential function f for the field $\mathbf{F} = 2xy\,\mathbf{i} + (x^2 - z^2)\,\mathbf{j} - 2yz\,\mathbf{k}$. **[A] / [CO5]**

- For writing find f such that $\nabla f = \mathbf{F}$ and $\dfrac{\partial f}{\partial x} = 2xy, \quad \dfrac{\partial f}{\partial y} = x^2 - z^2, \quad \dfrac{\partial f}{\partial z} = -2yz$ ... 2 Marks
- Calculations to find $f(x, y, z)$ using integration ... 3 Marks
- Final answer $f(x, y, z) = x^2y - yz^2 + C$. ... 1 Mark

Total Marks should not Exceed 6

Part C
(Essay Questions)
Answer any 3 questions
Each question carries 10 marks

Question 17 Let R_1 and R_2 be relations on a set A represented by the matrices **[An] / [CO2]**

$$M_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad M_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Find the matrices that represent

(a) $R_1 \cup R_2$

(b) $R_1 \cap R_2$

(c) $R_2 \circ R_1$

(d) $R_1 \circ R_2$

(e) $R_1 \oplus R_2$

- $M_{R_1 \cup R_2} == \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. ... 2 Marks
- $M_{R_1 \cap R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$. ... 2 Marks
- $M_{R_2 \circ R_1} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. ... 2 Marks

- $M_{R_1 \circ R_2} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. ... 2 Marks
- $M_{R_1 \oplus R_2} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. ... 2 Marks

Total Marks should not Exceed 10

Question 18 Show that $\bar{\mathbf{v}}(t)(\mathbb{Z}, \leq)$ is totally ordered, but not a well-ordered set. **[U] / [CO3]**

- For showing $(\mathbb{Z}, \leq)$ is totally ordered ... 5 Marks
- For showing $(\mathbb{Z}, \leq)$ is not a well-ordered set ... 5 Marks

Question 19 Let S be a set of strings of zeros and ones. For any positive integer n, define the relation R_n on S such that sR_nt if and only if $s = t$, or both s and t have at least n characters and the first n characters of s and t are the same. Show that R_n is an equivalence relation. Also find three elements equivalent to 0101 in R_4. **[U] / [CO3]**

- For showing R_n is reflexive ... 2 Marks
- For showing R_n is symmetric ... 2 Marks
- For showing R_n is transitive ... 3 Marks
- For writing any three elements equivalent to 0101 in R_4 ... 3 Marks

Question 20 Consider the curve defined by **[U] / [CO4]**

$$\mathbf{r}(t) = (1 + 2t)\mathbf{i} + (1 + 3t)\mathbf{j} + (6 - 6t)\mathbf{k}$$

(a) Find the velocity vector and its magnitude.

(b) Find the arc length parameter $s(t)$ along the curve from the point where $t = 0$ by evaluating the integral

$$s = \int_0^t |\bar{\mathbf{v}}(t)| dt.$$

(c) Use the above arc length parameter to find the length of the curve for $-1 \leq t \leq 0$.

- Velocity vector:

$$\bar{\mathbf{v}}(t) = \mathbf{r}\,'(t) = 2i + 3j - 6k.$$

... 2 Marks

Magnitude of the velocity:

$$|\bar{\mathbf{v}}(t)\| = \sqrt{2^2 + 3^2 + (-6)^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7.$$

... 2 Marks

- Arc length from $t = 0$ to t:

$$s(t) = \int_0^t |\bar{\mathbf{v}}(t)|\, dt = \int_0^t 7\, dt = 7t.$$

Thus $s(t) = 7t$ (with $s(0) = 0$). Equivalently $t = \dfrac{s}{7}$. ... 4 Marks

- Length of the curve for $-1 \leq t \leq 0$:

$$\text{Length} = \int_{-1}^0 |\bar{\mathbf{v}}(t)|\, dt = \int_{-1}^0 7\, dt = 7\big(0 - (-1)\big) = 7.$$

... 4 Marks

Total Marks should not Exceed 10

Question 21 Find the line integral

$$\int_C (x^2 + y^2) dy$$

along the path $C = C_1 \cup C_2$ consisting of the line segment from (0, 0) to (3, 0) followed by the segment from (3, 0) to (3, 3). **[An] / [CO5]**

- For C_1:

$$y = 0, \quad \Rightarrow dy = 0.$$

... 3 Marks

Hence,

$$\int_{C_1} (x^2 + y^2)\, dy = 0.$$

... 2 Marks

- For C_2:

$$x = 3, \quad y = t, \quad 0 \leq t \leq 3, \quad dy = dt.$$

... 3 Marks

Substitute into the integral:

$$\int_{C_2} (x^2 + y^2)\, dy = \int_0^3 (9 + t^2)\, dt. = 36$$

... 2 Marks

- Final Answer $\displaystyle\int_C (x^2 + y^2)\, dy = 36.$ ... 1 Mark

Total Marks should not Exceed 10

Question 22 Integrate $x + \sqrt{y}$ over the curve $C = C_1 \cup C_2$ where C_1 is the curve $y = x^2$ and C_2 is the straight line $y = x$.

- For taking $C_1 : \ x = t, \ y = t^2, \ 0 \leq t \leq 1.$ ... 2 Marks
  For finding $\displaystyle\int_{C_1} (x + \sqrt{y})\, ds = \frac{1}{6}\big(5^{3/2} - 1$ ... 3 Marks
- For taking $C_2 : \ x = t, \ y = t, \ 0 \leq t \leq 1.$ ... 2 Marks
  $\displaystyle\int_{C_2} (x + \sqrt{y})\, ds = \frac{7\sqrt{2}}{6}$ ... 2 Marks
- Final Answer $\displaystyle\int_{C} (x + \sqrt{y})\, ds = \frac{1}{6}\big(5\sqrt{5} - 1 + 7\sqrt{2}\big)$ ... 1 Mark

Give 10 marks for a reasonable attempt that correctly explains that the region is unbounded, and therefore, it is not possible to evaluate the integral.

Give 5 marks for any other reasonable attempt.